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Article

Damage-less observation of polymers by electron dose control in scanning electron microscope

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Abstract

Methodology for quantitative evaluation of electron radiation damage and calculation of tolerable electron dose was developed to achieve damage-less scanning electron microscope (SEM) observation of beam-sensitive polymer film. The radiation damage is typically evaluated with visual impressions of SEM images; however, this method may be unreliable because observer's subjectivity may affect the results. Evaluation with quantitative value is crucial to improve reliability. In this study, the radiation damage was evaluated by using normalized correlative coefficient (R_{NCC}) between an initial frame and latter frames of the multiple SEM images that were taken consecutively. Tolerable dose was obtained by defining a threshold point of R_{NCC} where rapid reduction of R_{NCC} started. A SEM image with less damage and acceptable signal-to-noise ratio was obtained by integrating the images from the initial frame to the tolerable frame.

Key words: polymer, lithium ion battery, separator, damage-less observation, ultra-low voltage scanning electron microscope, dose control

Introduction

Field emission scanning electron microscopes (FE-SEMs) are widely used to observe fine structures of various types of materials. In the development of lithium-ion battery (LIB), which has been an essential power supplier in our life, FE-SEM is utilized as an observation tool for fine structures of materials in LIB [1]. Separator film, which is porous polymer film mounted between an anode and a cathode, is critical for performance and safety of LIB. Lithium ions can transport through the separator film via electrolyte, while the anode and cathode are electrically isolated. Another role of the separator film is a safety system. When the temperature increases abnormally, the pores in the separator film are melted and stop the lithium-ion transport. These properties of the separator film are dependent on the pore size and the surface morphology; thus, the evaluation of the fine structures using FE-SEM is crucial.

Examples of SEM images of the separator film are shown in Fig. 1. In Fig. 1a, fibril structures, pores and the surface morphology on the base structures are clearly observed. Polymers are commonly sensitive to electron beam irradiation and can be easily degraded (radiation damage) as shown in Fig. 1b. Accurate evaluation of the materials is difficult if the structures are degraded. The radiation damage is typically evaluated with visual impressions of SEM images; however, this method may be unreliable because observer's subjectivity may affect the results. Evaluation with quantitative value is crucial to improve reliability.

The radiation damages of polymers observation in SEM are mainly caused by inelastic scattering of primary electrons, which

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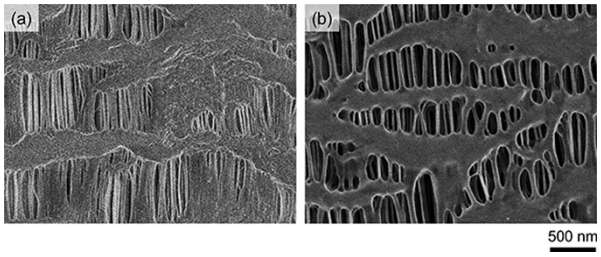


Fig. 1. SEM images of the LiB separator with (a) less radiation damage and (b) radiation damage ($V_i = 50$ V).

is categorized into two mechanisms [2–4]. One is heating damage caused by a temperature rise due to energy transfer from the primary electrons to the specimen. Polymers are generally sensitive to heating due to their poor thermal conductivity and low melting point. The heating damage can be mitigated by cooling the specimen. The other is ionization caused by radiation, which directly leads to the destruction of chemical bonds. This ionization leads to cross-linking and decomposition, and the damage diffuses subsequently [4]. Although the destruction of chemical bonds due to the ionization still occurs at low temperature, the subsequent diffusion of the damage can be reduced by cooling the specimen [4]. Reduction of the electron dose is effective to reduce the amount of the radiation damage due to the ionization [2–4]. However, signal-to-noise ratio (SNR) decreases, and the evaluation of the structures becomes difficult, when the electron dose is reduced. A high dose is desired as long as the radiation damage does not occur (a tolerable dose).

In this study, we propose methodology for quantitative evaluation of electron radiation damage and calculation of tolerable electron dose. We also discuss the mechanism of the radiation damage on the separator film.

Methods

Methodology of the dose control

The methodology of the dose control is an application of frame integration, which is generally used to improve SNR of SEM images. Multiple frames of SEM images as elemental frames for the image integration are taken consecutively. Dose rate (a dose condition of each single frame) D_r is defined as

$$D_r = \frac{I_p \cdot t}{S} \tag{1}$$

where I_p is probe current, t is dwell time and S is pixel area. Accumulated electron dose at each frame is obtained as

$$D = D_r \cdot N \tag{2}$$

where N is a frame number. The SEM image with a predetermined dose condition is obtained by integrating from an initial frame to the frame whose accumulated dose corresponds to the predetermined dose. SEM image with different accumulated dose can be obtained later by reselecting the integrated frames. As mentioned above, there are two types of radiation damage in SEM observation; however, the SEM image can only show the morphological changes appearing as a result of the radiation damage. Thus, the radiation damage is evaluated through the morphological changes in this method. The degradation of the surface structures by the beam irradiation was evaluated by using a normalized correlation coefficient (R_{NCC}) between the initial frame and each latter frame. R_{NCC} is defined as

$$R_{\mathrm{NCC}} = \frac{\sum \left(f_{(i,j)} - f_m\right)\left(T_{(i,j)} - T_m\right)}{\sqrt{\sum \left(f_{(i,j)} - f_m\right)^2}\sqrt{\sum \left(T_{(i,j)} - T_m\right)^2}} \tag{3}$$

where $f_{(i,j)}$ and $T_{(i,j)}$ are pixel gray levels at the coordination of (i, j) of a reference image and an evaluated image, f_m and T_m are average gray levels of the reference image and the evaluated image. In this experiment, maximum value of R_{NCC} is normalized as 1. R_{NCC} is a value that represents similarity between two different images, and it is applied to pattern-matching process [5]. If the two images are similar, R_{NCC} shows a higher value up to 1, and if the two images are different, R_{NCC} shows a smaller value. If the images are monochrome inversion, R_{NCC} shows -1. In this experiment, the initial frame was considered as the reference image, and R_{NCC} between the initial and the latter frames were calculated and plotted. When the frames and the accumulated electron dose increase, the degradation of the structure progresses and the R_{NCC} decreases. A tolerable frame that preserves the original structure is obtained by defining a threshold point where rapid reduction of R_{NCC} started. A tolerable dose is calculated by multiplying D_r by the number of the tolerable frame. Although SNR of each single frame is very low because of low D_r, a less-damaged image with acceptable SNR is obtained by integrating the frames from the initial to the tolerable frame.

Instrument

A Schottky-type FE-SEM (Hitachi SU7000) was used in this study. In this microscope, positive bias voltage is applied to a liner electrode installed inside the column for beam acceleration. This optics can reduce chromatic aberration significantly and allow high-resolution imaging even under ultra-low energy condition [6].

Specimen

A piece of uncoated separator film of the LiB was used as a specimen. The separator film is shown in Fig. 1, and it is made of polyolefin.

Experimental methods

Damage-less observation of the separator film was achieved by the dose–control method. The specimen has fine structure on the surface as can be confirmed in Fig. 1a. The landing energy should be lower to observe the fine surface structure of organic materials such as polymers since the interaction volume of the primary electrons can be suppressed. The landing energy of 50 eV may be the most suitable to observe the topmost surface features, because the inelastic cross-section generally becomes minimum at 50 eV [7]. V_i, I_p and D_r were set at 50 V, 5 pA and 1.2×10^{-6} C/cm^2, respectively. Totally 32 frames were obtained. The threshold point of R_{NCC} was defined at the point where rapid reduction of R_{NCC} started. The structure dependence on the radiation damage was also examined. The specimen has both fibril and base structures, and R_{NCC} of those areas were evaluated. The effect of D_r was also examined with D_r of 7.4×10^{-7}, 1.2×10^{-6} and 2.5×10^{-6} C/cm^2. Totally 60 frames were obtained. Time interval experiments were also conducted to clarify the damage mechanism. If the heating damage is dominant, heat accumulation should be reduced by placing the time intervals between frames. SEM images were taken with the 10-s intervals between frames, and the attenuation of R_{NCC} was investigated. The effect of V_i on the tolerable dose was also examined. The tolerable

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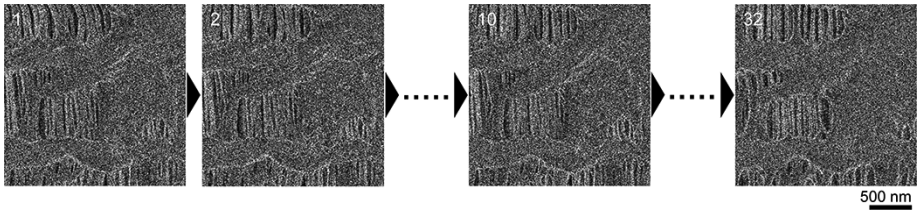


Fig. 2. The 1st, 2nd, 10th and the 32nd frames of the series of SEM images (Vi = 50 V, Dose rate = 1.2×10^{-6} C/cm^2).

doses and the integrated images were compared at $V_i = 10$, 30, 50, 100, 200, 500, 1000, 3000 V.

Results

Damage-less observation by dose control

Multiple frames of the SEM images were taken consecutively at $D_r = 1.2 \times 10^{-6}$ C/cm^2. Figure 2 shows the 1st, 2nd, 10th and the 32nd frame of the series of the SEM images. The upper edge of the first frame may be distorted due to beam drifting caused by charging, thus the first frame was excluded and the second frame was considered as the initial frame. SNR of each single frame without image integration may be too low to evaluate the degradation manually with human eyes. R_{NCC} between the initial frame and other latter frames were plotted against the number of frames in Fig. 3. R_{NCC} showed a value close to 1 while the degradation was small until the 11th frame, and the rapid reduction of R_{NCC} started after the 12th frame. Thus, the tolerable frame was determined as the 11th frame and the threshold of R_{NCC} was defined as 0.85. The tolerable dose was obtained as 1.4×10^{-5} C/cm^2. Figure 4a shows an integrated image from the initial frame to the 11th frame (the accumulated dose was 1.4×10^{-5} C/cm^2), Fig. 4b shows an integrated image from 23rd frame to 32nd frame (the accumulated dose was 4.0×10^{-5} C/cm^2) and Fig. 4c shows an atomic force microscope (AFM) image of the same specimen as a reference. Thin fibril structures (shown by arrows) and fine surface morphologies (shown by arrow heads) of the specimen are clearly observed in Fig. 4a, which are similar to the AFM image in Fig. 4c. Those structures were lost, and the number of the fibrils decreases, and the thickness of the fibrils becomes larger by the radiation damage in Fig. 4b. However, the fibrils were not cut by the radiation damage.

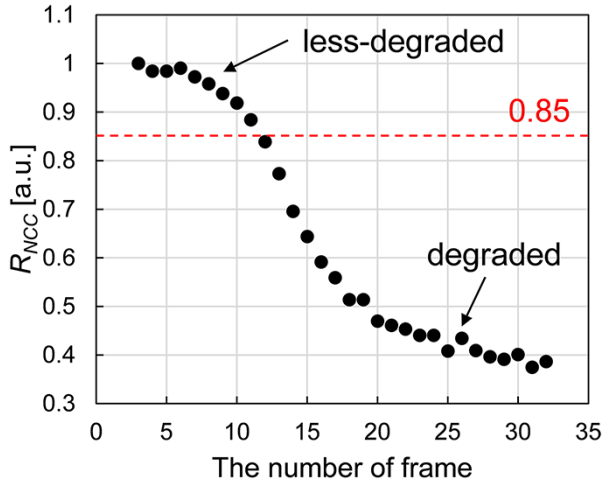


Fig. 3. R_{NCC} between the initial frame and other latter frames plotted against the number of frames.

R_{NCC} of the fibril and base structures were evaluated to investigate the structure dependence on the radiation damage. Figure 5 shows the SEM images of the selected area of the fibril structure, and R_{NCC} was plotted against the number of the frames. Figure 6 shows the SEM images of the base structure and the variation of R_{NCC}. The trend of the R_{NCC} variation of the fibril area is similar to that of the whole area shown in Fig. 3, and R_{NCC} shows higher value until the 12th frame. On the other hand, R_{NCC} of the base structure shows linear decrease. The attenuation of R_{NCC} was different between the

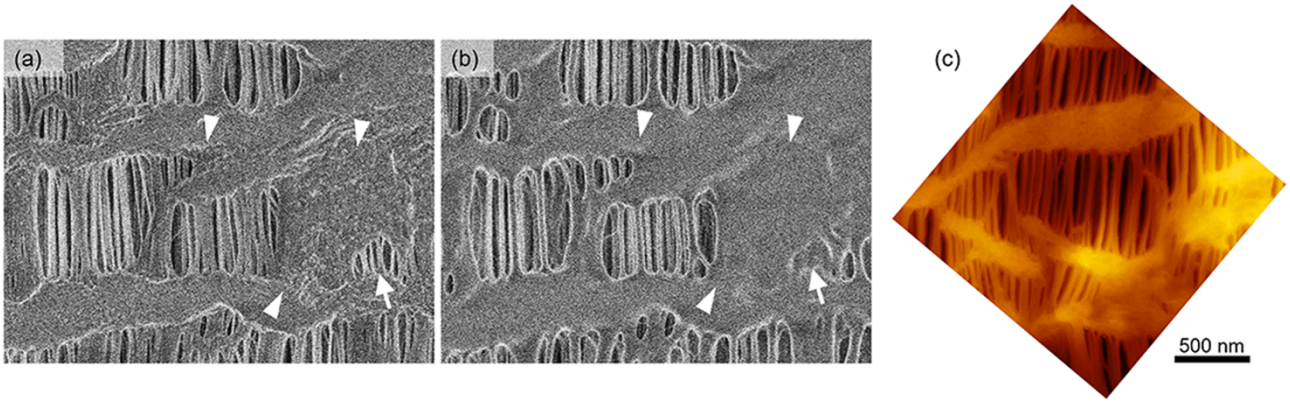


Fig. 4. (a) Integrated image from the initial frame to the tolerable frame (11th frame) ($V_i = 50$ V, accumulated dose is 1.4×10^{-5} C/cm^2). (b) Integrated image from 23rd frame to 32nd frame ($V_i = 50$ V, accumulated dose is 4.0×10^{-5} C/cm^2). (c) AFM image of the same specimen.

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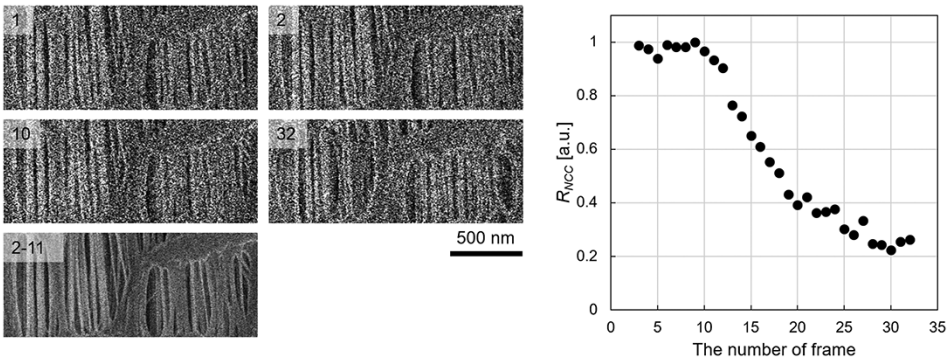


Fig. 5. SEM images of the selected area of the fibril structure and the variation of R_{NCC} against the number of the frames.

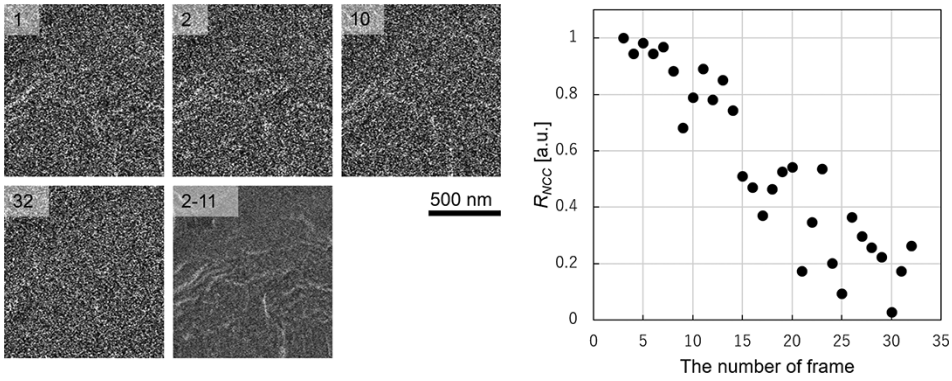


Fig. 6. SEM images of the selected area of the base structure and the variation of R_{NCC} against the number of the frames.

fibril structure and the base structure; that is, the base structure is degraded faster than the fibril structure.

The variations of R_{NCC} obtained at different D_r conditions were compared to investigate the effect of D_r. Figure 7 shows the variations of R_{NCC} at $D = 7.4 \times 10^{-7}$, 1.2×10^{-6}, 2.5×10^{-6} C/cm^2. The trends of the R_{NCC} variations at $D = 7.4 \times 10^{-7}$, 1.2×10^{-6} C/cm^2 were almost similar. However, the variation of R_{NCC} at $D = 2.5 \times 10^{-6}$ C/cm^2 is slightly shifted. It suggests that too high D_r may not work, because the radiation damage starts even at the initial frame. R_{NCC} is not suitable for the degradation evaluation when the initial frame is already degraded, because R_{NCC} is based on the initial frame.

R_{NCC} was obtained with the 10-s interval between frames to investigate the radiation damage. Figure 8 shows a comparison of R_{NCC} variations between with and without the 10 second interval. The tendencies of the attenuation of R_{NCC} with and without interval were almost the same.

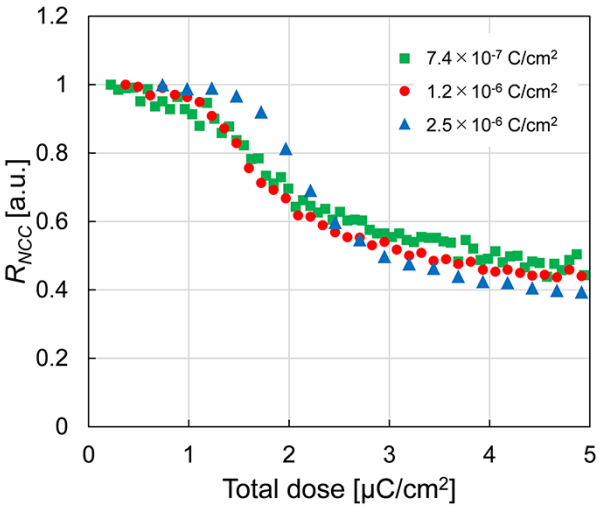


Fig. 7. The variation of R_{NCC} at $D_r = 7.4 \times 10^{-7}$, 1.2×10^{-6}, 2.5×10^{-6} C/cm^2.

Landing energy dependence

The tolerable doses at V_i from 10 to 3000 V were determined from Fig. 9, which is the variation of R_{NCC} against the accumulated dose. At $V_i = 10$–200 V, the trends of the R_{NCC} variation were similar and the tolerable doses were obtained as 1.4×10^{-5} to 1.5×10^{-5} C/cm^2. At $V_i > 200$ V, the higher the V_i is, the gentler the reduction of R_{NCC}. The tolerable doses at each V_i are plotted in Fig. 10. The tolerable doses were gradually increased at $V_i > 500$ V. The tolerable dose at $V_i = 3$ kV was obtained as 4.4×10^{-5} C/cm^2, which is approximately three times higher than those at $V_i = 10$–200 V. Thus, the degradation caused by the beam irradiation tends to be larger at lower V_i than higher V_i in this experiment. The integrated images obtained within each tolerable dose are shown in Fig. 11. At least one area of very thin fibril structure, which will be easily lost at a high-dose condition, was observed in each image as shown by arrows.

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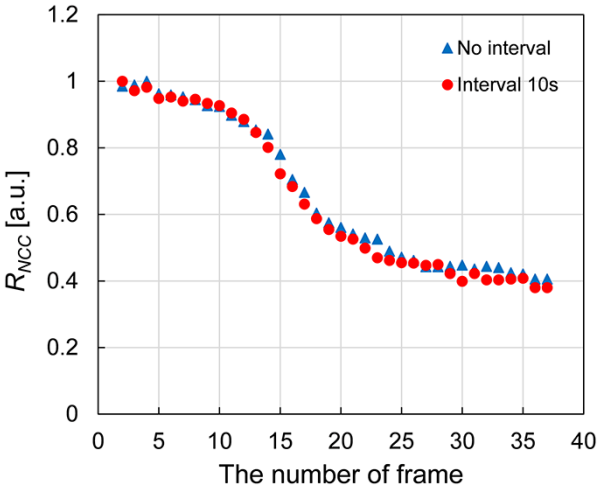


Fig. 8. Comparison of R_{NCC} variations between with and without the 10-s interval.

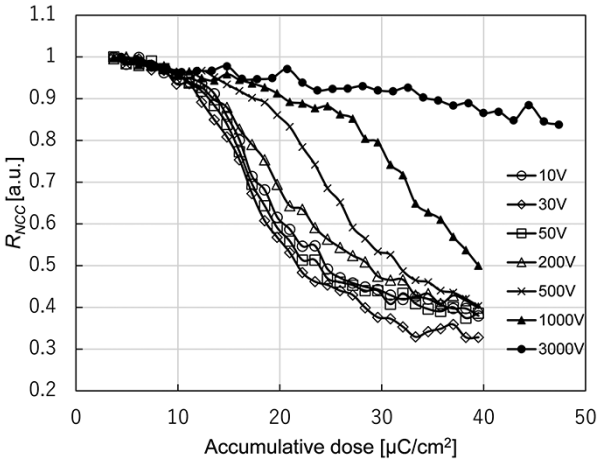


Fig. 9. Variation of R_{NCC} plotted against the accumulated dose at $V_i = 10$, 30, 50, 100, 200, 500, 1000, 3000 V.

Discussion

The radiation damage is typically evaluated with visual impressions of SEM images; however, this method may be unreliable because observer's subjectivity may affect the results. Evaluation with quantitative value is crucial to improve reliability. In this study, the radiation damage was evaluated by using R_{NCC} between an initial frame and latter frames of the multiple SEM images. In TEM observation, 'a critical dose' was defined for complete fading of an electron diffraction pattern of an organic crystalline specimen [2]. However, it is difficult to apply this method to the SEM observation, because the electron diffraction pattern is not available. In this paper, the critical dose (called as the tolerable dose in this paper) was obtained using the variation of R_{NCC}. The degradation can be reduced by limiting the dose within the tolerable dose as shown in Fig. 3. Even if the SNR is unacceptable, the electron dose cannot be raised over this tolerable dose. Thus, the improvement of detector of efficiency is desired.

As mentioned above, there are two types of the radiation damage in the SEM observation: heating and ionization. If the heating damage is dominant, the heat accumulation should be reduced by placing the time intervals between frames. However, there was no difference in the attenuation of R_{NCC} between with and without the 10-s interval in the experiment shown in Fig. 8. The temperature rise ΔT in the case of a stationary probe can be estimated as

$$\Delta T = \frac{1.5}{\pi} \frac{I_p \cdot V_i}{\lambda \cdot R} \tag{4}$$

where λ is thermal conductivity (W/m/K) and R is the electron range (μm) [2,3]. The temperature rise is proportional to the probe current and the landing energy and inversely proportional to the thermal conductivity and the electron range. Under this experimental condition ($I_p = 5$ pA, $V_i = 50$ eV, $\lambda = 0.1$ W/m/K, R = 0.5 nm), ΔT was estimated as 3K. In the case of the scanning probe, ΔT may be lower because the pixel dwell time may be sufficiently shorter than the thermal equilibrium time [2,3]. These results indicate that the ionization is more dominant rather than heating under this experimental condition. When I_p becomes higher, it may be necessary to consider heating damage.

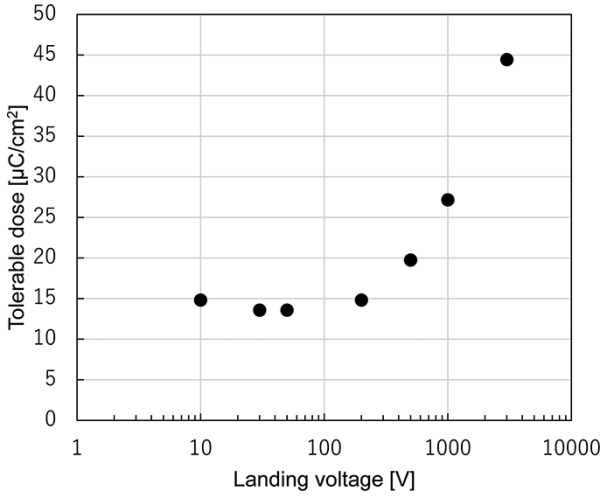


Fig. 10. Tolerable dose plotted against V_i.

As shown in Figs. 5 and 6, the trends of the attenuation of R_{NCC} were different between the fibril structure and the base structure; the base structure is degraded faster than the fibril structure. As shown in Fig. 4a and b, the number of the fibers decreased and the thickness of the fibers increased, but the fibrils were not cut, by the radiation damage. Thus, the damage progresses in two steps. As the first step, the topmost surface of both base and fibril structures may be melted and become fluid by the radiation. The morphological change at this step is only the loss of fine surface structure. The attenuation of R_{NCC} in the fibril structure is small, since the amount of the fine surface structure on the fibril is small. On the other hand, R_{NCC} in the base structure decreased linearly, since there originally exists the fine surface structure on the base structure. As the second step, R_{NCC} in the fibril structure started to decrease because of the decrement of the number of the fibrils and the increment of the diameter of the fibrils. These structural changes may be caused by the surface tension, which leads to cohesion of the fibrils.

The experimental results show that the radiation damage occurs more rapidly at lower voltage. One of the causes may be that the energy density deposited per unit volume is greater at lower voltage.

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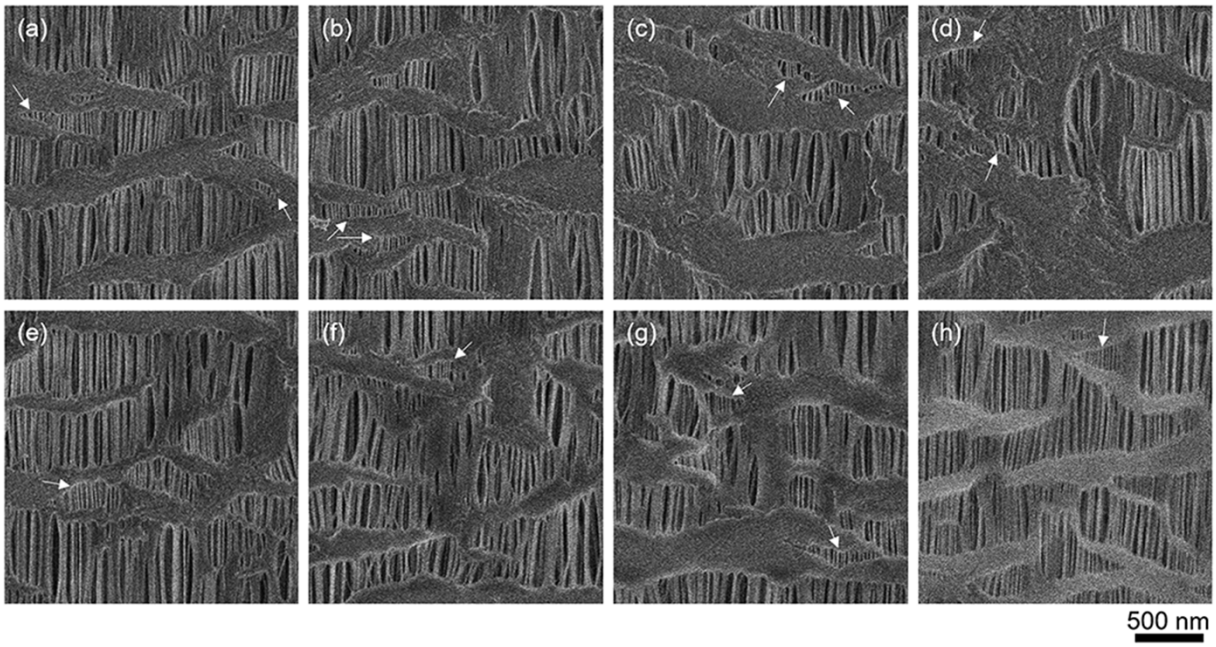


Fig. 11. Comparison of integrated images obtained within each tolerable dose at V_i = (a) 10 V, (b) 30 V, (c) 50 V, (d) 100 V, (e) 200 V, (f) 500 V, (g) 1 kV, (h) 3 kV.

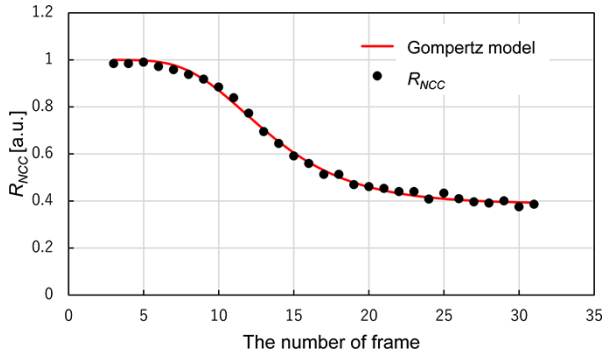


Fig. 12. The fitting result of the variation of R_{NCC} to Eq. (4).

This fact agrees the prediction reported by Joy and Joy [8]. Therefore, the accumulated dose may not be a good factor of the damage vs landing energy. In the experimental results shown in Fig. 10, the tolerable dose at $V_i = 3$ kV was approximately three times higher than that at $V_i = 50$ V. Since the interaction volume is proportional to E^5 by rough calculation [8], the interaction volume at 3 kV is 60^5 times larger than that at 50 V. Therefore, the dose density, which divides the electron dose by the interaction volume, is not appropriate. Complicated phenomena including charging may also affect the damage evolution at the higher landing energy.

Joy and Joy [8] also predicted that the radiation damage might be completely eliminated if the SEM observation could be operated at extremely low energy, 25 eV or less, because the inelastic scattering cross-section becomes lower at that condition. In this experiment, the V_i was reduced down to 10 V; however, the radiation damage still occurred. The tendency of the inelastic scattering cross-section in this energy range is different depending on the materials. The inelastic scattering cross-section of this separator film may not be reduced in this energy range.

The variation of R_{NCC} shows an S-shaped curve. It is known that this kind of the S-shaped curve can be fit to the sigmoid equation, and the Gompertz model is one of the equations [9]. The fitting result of the variation of R_{NCC} to an equation that is based on Gompertz model is shown in Fig. 12. The fitting equation is defined as

$$R_{NCC}(N) = 1 - ab^{c^N} \tag{5}$$

where the values of a, b, c are fitting parameters. The Gompertz model has been widely used for modeling growth curves, such as animal population. Recently, it has been also applied to 'K indicator', which has been suggested to represent and predict the growth status in coronavirus disease 2019 (COVID-19) spread [10]. The variation of R_{NCC} was well fit to the Gompertz model as shown in Fig. 12. The decreasing behavior of R_{NCC} might be represented as this growth curve.

The preset dose control methodology can be applied to other phenomena caused by the accumulation of the electron dose, such as controlling charging effect and enhancing voltage contrast. However, these phenomena may be more complicated than the radiation damage because both accumulation and relaxation of charging need to be considered. Since the proposed methodology is based on the image, the evaluation may be disturbed when the charging takes place. Especially at higher voltage, abnormal contrast and image distortion may occur due to charging.

The methodology for quantitative evaluation of electron beam damage may be applicable not only for the LiB separator but also for other beam-sensitive materials such as photoresist patterns, because pattern shrinkage may also affect R_{NCC}. Edge contrast of the pattern may be clearly observed in a secondary electron image, and it may enhance the change of R_{NCC}.

In this research, we have successfully demonstrated the methodology for quantitative evaluation of electron radiation damage

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and calculation of tolerable electron dose. We hope that this methodology helps in the accurate evaluation of surface structure of polymer and battery materials, which are sensitive to electron beam irradiation, and that leads to improve reliability of the SEM observation.

Concluding remarks

Damage-less observation of weak polymer film was demonstrated by the dose control technique. The degradation of the surface structures by the beam irradiation was evaluated using the normalized correlation coefficient between the initial frame and the latter frames. The tolerable dose was obtained by defining the threshold point where rapid reduction of the normalized correlation coefficient started. The less-degraded image with acceptable SNR was obtained by integrating the frames from the initial to the tolerable frame. The electron energy dependence on the tolerable dose was also examined. The radiation damage was reduced at higher landing energy condition than 1 keV.

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